MIL-E-1/121D
4 December 1972
SUPERSEDING
MIL-E-1/121C
4 March 1969

MILITARY SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING

TYPE 6BN6

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Pentode, miniature, gated beam

Outline	---	5-3 (EIA)
Base	---	E7-1
Envelope	---	T5-1/2
Cathode	---	Coated unipotential

Base connections:

Pin No.	---	1	2	3	4	5	6	7
Element	---	k	g1	h	h	g2 (accelerator grid)	g3	a

ABSOLUTE-MAXIMUM RATINGS:

(D) Parameter:	Ef	Eb	Ec1	eg1	Ec2	Ecc2	Ec3	Ik	Ehk	Rg2	Cg2	Alt
Unit:	V	Vdc	Vdc	v	Vdc	Vdc	Vdc	mAdc	v	Meg	μF	ft
Maximum:	6.9	165	---	60 (Note 2)	110	---	---	12.6	100	---	---	Note 1
Minimum:	5.7	---	---	---	---	---	---	---	---	---	---	---
(D) TEST CONDITIONS:	6.3	135	0	---	---	280	4.0	---	---	0.033	1.0	---

GENERAL:

Qualification - Required

(D) denotes changes

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METHOD	REQUIREMENT OR TEST	SAMPLE	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1								
(D) 1266	Total grid current (1)	3	Ec1 = 0; Ecc3 = -2 Vdc; Rg3 = 0.25 Meg	0.65	II	Ic3	---	-1.0	µAdc
(D) 1266	Total grid current (2)	6	Ec1 = Rb = 2 mAdc	0.65	II	Ic1	0	-0.05	µAdc
1256	Electrode current (1) (anode)	-		0.65	II	Ib	3.0	7.0	mAdc
1201	Short and discontinuity detection	-		0.4	II	---	---	---	---
	Quality conformance inspection, part 2								
1301	Heater current	-		---	---	If	275	325	mA
1331	Direct-interelectrode capacitance	-	No shield; g3 to all No shield; g1 to g3 No shield; g1 to all	6.5	Code E	C C C	2.7 --- 3.7	4.1 0.005 5.5	pF pF pF
1326	Heater-cathode leakage	-		---	---	Ihk	---	20	µAdc
(D) 1256	Electrode current (accelerator)	-	Ec2 = 75 Vdc; Rg2 = 0; Cg2 = 0	---	---	Ic2	2.5	6.5	mAdc
1256	Electrode current (2) (anode)	-	Ef = 5.5 V	2.5	I	Ib	2.7	---	mAdc
(D) 1256	Electrode current (3) (anode)	-	Ec1 = -7 Vdc; Ec3 = 0; Rb = 0.1 Meg; Ebb = 135 Vdc	2.5	I	Ib	---	20	µAdc
(D) 1256	Electrode current (4) (anode)	-	Ec1 = 0; Ec3 = -15 Vdc; Rb = 0.1 Meg; Ebb = 135 Vdc	---	---	Ib	---	20	µAdc
1306	Transconductance (1)	-	Ec3 = 0	2.5	I	Sm (g1-a)	80	640	µmhos
1306	Transconductance (2)	-	Ec3 = 0	2.5	I	Sm (g3-a)	350	1,200	µmhos
(D) 1211	Insulation of electrodes	-		4.0	S3	---	---	---	---
1121	Base strain	5		---	---	---	---	---	---
2126	Glass strain	-		2.5	I	---	---	---	---
1105	Permanence of marking	-		---	---	---	---	---	---
	Quality conformance inspection, part 3								
1501	Intermittent life	4	Group A; Eb = Ec2 = 100 Vdc; Eg1 = 15 Vac; Eg3 = 5 Vac; F = 60 Hz; Ehk = -180 Vdc; Rk = 200 ohms	---	---	---	---	---	---

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METHOD	REQUIREMENT OR TEST	[illegible]	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 3 -Continued								
---	Intermittent life-test end point: (500 hours)	-							
1256	Electrode current (1) (anode)	-		---	---	Ib	2.5	---	mAdc

NOTES:

1. See "Reduced pressure (altitude) rating", and altitude, maximum peak voltage in the basic document.
2. Peak positive grid No. 1 voltage.
3. This test to be performed at the conclusion of the holding period.
4. Eg1 is in phase with Eg3.
5. The "Base-strain test, miniature tube, sampling" paragraph of MIL-E-1 and all subparagraphs thereto (Normal inspection, Sampling, Acceptance and rejection criteria, Rejected tubes, and Records) shall apply to this test.
6. Prior to this test the tubes shall be preheated for a minimum of 5 minutes under the test conditions.

Custodians:
Army - EL
Navy - EC
Air Force - 80

Review activity:
DSA - ES

User activities:
Army - WC
Navy - AS, OS, MC, CG, SH
Air Force - 11, 19

Preparing activity: Navy - EC

Agent: DSA - ES

(Project 5960-2709)

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NAVAL ELECTRONIC SYSTEMS COMMAND
WASHINGTON, D. C. 20360

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OMB Approval
No. 22-R255

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A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 68 WHICH MAY BE USED S/N 0102-014-1802

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